Calculus Homework – Function series and power series

Exercise 1. *Determine the set of convergence of the following function series!*

$$(1.1)\ \sum_{n=1}^{\infty}\left(\frac{2x+4}{5x+2}\right)^n \qquad (1.2)\ \sum_{n=1}^{\infty}\left(\frac{x+1}{x^2+2}\right)^n \qquad (1.3)\ \sum_{n=1}^{\infty}\left(\frac{3x-6}{x+7}\right)^n$$

Exercise 2. *Calculate the radius of convergence of the following power series! Determine their set of convergence as well!*

$$(2.1)\ \sum_{n=0}^{\infty}\frac{n!}{n^n}(x+2)^n \qquad (2.2)\ \sum_{n=0}^{\infty}\frac{\cos(n\pi)}{3^n(n+2)^2}(x-1)^n \qquad (2.3)\ \sum_{n=0}^{\infty}\frac{(3x)^n}{(n+1)!}$$

$$(2.4)\ \sum_{n=0}^{\infty}\frac{5^{n+1}}{n+3}(x+1)^n \qquad (2.5)\ \sum_{n=0}^{\infty}\frac{(n^2+2)x^n}{(n+2)!} \qquad (2.6)\ \sum_{n=0}^{\infty}(3-x)^n$$

$$(2.7)\ \sum_{n=0}^{\infty}\left(\frac{-2x}{n+1}\right)^n \qquad (2.8)\ \sum_{n=0}^{\infty}\left(\frac{n+2}{n+1}\right)^{n^2}(x+3)^{n+1} \qquad (2.9)\ \sum_{n=0}^{\infty}\frac{x^n}{2^n}$$

Exercise 3. *Calculate the radius of convergence of the following power series! Determine their set of convergence as well!*

$$(3.1)\ \exp(x) := \sum_{n=0}^{\infty}\frac{x^n}{n!}$$

$$(3.2)\ \sin(x) := \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n+1}}{(2n+1)!} \qquad (3.3)\ \cos(x) := \sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}$$

$$(3.4)\ \sinh(x) := \sum_{n=0}^{\infty}\frac{x^{2n+1}}{(2n+1)!} \qquad (3.5)\ \cosh(x) := \sum_{n=0}^{\infty}\frac{x^{2n}}{(2n)!}$$

Look for the graph of the above functions!